JR Validated Environment — Correlation Analysis Module

User Guide for Correlation and Regression Analysis

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1. How to Use This Guide

This guide describes the four correlation and regression scripts in the JR Validated Environment. Use the three routing questions below to find the right script, then go to the per-script reference in Section 4.

Three routing questions

Question		Guidance
Are you comparing two measurement methods (e.g. two instruments, two operators, two assays)?	Yes →	Use jrc_corr_passing_bablok (Section 4.4). Also consider running jrc_bland_altman from the community scripts for a complementary agreement analysis.
Do you want a fitted line (y = mx + b) with R^2 and residuals?	Yes →	Use jrc_corr_regression (Section 4.3).
Is your data non-normal, ordinal, or skewed?	Yes →	Use jrc_corr_spearman (Section 4.2). No normality assumption required.
None of the above — you just want to measure linear association with a CI.	→	Use jrc_corr_pearson (Section 4.1). Standard choice for continuous normal data.

NOTE: All four scripts accept the same CSV format: two numeric columns. The default column names are x and y. Use --xcol and --ycol to specify different column names if your file uses other headers.

2. Quick Reference Table

Find the row that matches what you are trying to do, then go to the section listed.

I want to…	Script	Section
Measure linear association with a 95% CI	jrc_corr_pearson	4.1
Measure monotonic association without assuming normality	jrc_corr_spearman	4.2
Fit a line and get R^2, slope, intercept, residual diagnostics	jrc_corr_regression	4.3
Compare two measurement methods (no gold standard required)	jrc_corr_passing_bablok	4.4

3. Understanding the Results

3.1 Pearson r and Spearman ρ (rho)

Both statistics measure association between −1 and +1. The sign tells you direction (positive: both rise together; negative: one rises as the other falls). The magnitude tells you strength: |r| or |ρ| < 0.3 is weak, 0.3–0.7 is moderate, > 0.7 is strong. A p-value < 0.05 means the association is statistically significant at the 5% level.

IMPORTANT: A significant p-value does not confirm practical importance. With large n, even a very small r can be significant. Always read the magnitude alongside the p-value, and consider whether the CI is narrow enough to be useful.

3.2 Confidence Interval for Pearson r

The 95% CI (shown as [lower, upper]) gives the range of plausible values for the true population correlation. A CI that does not include 0 confirms significance. A wide CI (e.g., [0.10, 0.85]) means the estimate is imprecise — collect more data. Spearman rho does not have a CI in this script; use Pearson if a CI is required.

3.3 R^2 (Regression)

R^2 is the proportion of variation in y explained by the linear model. $R^2 = 0.90$ means 90% of y's variation is accounted for by x. The residual plot (right panel) should show points scattered randomly around zero with no systematic pattern — a curved or funnel shape suggests the linear model is not appropriate.

3.4 Passing-Bablok Slope and Intercept

The key question is whether the two methods are interchangeable. Two criteria must hold:

- Slope CI includes 1: no proportional difference (one method does not read systematically higher than the other by a factor).
- Intercept CI includes 0: no constant offset (one method does not read systematically higher by a fixed amount across the whole range).

If either CI excludes its target value, the methods are not interchangeable and a conversion factor or offset correction is needed. The Cusum test checks whether the relationship is linear across the measurement range — a rejected Cusum means the P-B line may not be a good summary of the relationship.

TIP: Run jrc_bland_altman (community scripts) alongside jrc_corr_passing_bablok for a complete method comparison: Passing-Bablok gives you the regression line and interchangeability tests; Bland-Altman shows the magnitude and distribution of differences at each level of measurement.

4. Per-Script Reference

4.1 jrc_corr_pearson — Pearson Correlation

When do I use this?

Use when both variables are continuous and approximately normally distributed, and you want to quantify the strength and direction of their linear association with a confidence interval. This is the standard first choice for continuous measurement data.

What do I need?

Argument	What to provide
data.csv	CSV with two numeric columns. Default column names: x and y. Minimum 3 complete (non-missing) paired observations.
--xcol <name>	Optional. Name of the x column if different from 'x'.
--ycol <name>	Optional. Name of the y column if different from 'y'.
--conf <value>	Optional. Confidence level for the CI (default: 0.95).

Example

```
jrc_corr_pearson ~/data/study_results.csv --xcol force_N --ycol displacement_mm
```

What to look for in the output

Look for Pearson r and its CI. A CI that does not cross 0 confirms a real association. The strength label (weak / moderate / strong) is printed below the statistics. A scatter plot with best-fit line is saved to ~/Downloads/ — check that the relationship looks linear; if the points follow a curve, Pearson r understates the true association and jrc_corr_regression or jrc_corr_spearman may be more appropriate.

4.2 jrc_corr_spearman — Spearman Rank Correlation

When do I use this?

Use when your data may not be normally distributed, contains outliers, is measured on an ordinal scale, or when you want a non-parametric alternative to Pearson. Spearman measures monotonic association (does one variable tend to increase as the other increases?) without requiring linearity.

What do I need?

Argument	What to provide
data.csv	CSV with two numeric columns. Default column names: x and y. Minimum 3 complete (non-missing) paired observations.
--xcol <name>	Optional. Name of the x column if different from 'x'.
--ycol <name>	Optional. Name of the y column if different from 'y'.

Example

```
jrc_corr_spearman ~/data/study_results.csv --xcol torque_Nm --ycol
cycles_to_failure
```

What to look for in the output

The output shows Spearman ρ (rho) and a p-value. Interpret the magnitude the same way as Pearson r: < 0.3 weak, 0.3–0.7 moderate, > 0.7 strong. Note that no CI is provided for Spearman ρ — use jrc_corr_pearson if a CI is required and normality is reasonable. The scatter plot shows ranked x vs ranked y — a monotonically increasing pattern confirms a positive relationship.

NOTE: Spearman rho does not produce a confidence interval. If your design verification protocol requires a CI on the association measure, either justify the use of Pearson r (with a normality check) or collect enough data that the p-value alone is sufficient evidence.

4.3 jrc_corr_regression — Simple Linear Regression

When do I use this?

Use when you need a calibration line, want to quantify how much y changes per unit of x (the slope), or need R^2 to show what fraction of y's variation is explained by x. OLS regression assumes x is measured without error and that the relationship is linear — if both variables have measurement error and there is no gold-standard reference, use jrc_corr_passing_bablok instead.

What do I need?

Argument	What to provide
data.csv	CSV with two numeric columns. Default column names: x and y. Minimum 3 complete (non-missing) paired observations.
--xcol <name>	Optional. Name of the x column (predictor, x-axis).
--ycol <name>	Optional. Name of the y column (response, y-axis).
--conf <value>	Optional. Confidence level for slope and intercept CIs (default: 0.95).

Example

```
jrc_corr_regression ~/data/calibration.csv --xcol reference_kPa --ycol
sensor_kPa
```

What to look for in the output

Check the slope and intercept with their CIs: if the slope CI includes 0 the predictor is not useful. R^2 tells you how well the line fits — $R^2 > 0.90$ is typically required for a calibration relationship. The left panel of the PNG shows the scatter with the fitted line and 95% confidence band; the right panel shows residuals vs fitted values — residuals should scatter randomly around zero. A funnel shape (increasing spread with fitted value) means variance is not constant and the model assumptions are violated.

4.4 jrc_corr_passing_bablok — Passing-Bablok Regression

When do I use this?

Use when you are comparing two measurement methods and neither is a definitive gold standard. Unlike OLS regression (which assumes x is error-free), Passing-Bablok treats both variables symmetrically — it does not matter which instrument you label x and which you label y. Typical use: comparing a new instrument to an existing one, or comparing two assay methods before claiming equivalence.

What do I need?

Argument	What to provide
data.csv	CSV with two numeric columns. Each row is one specimen measured by both methods. Default column names: x and y. Minimum 3 paired observations; 20+ recommended for stable CI estimates.
--xcol <name>	Optional. Column name for method 1 (x-axis).
--ycol <name>	Optional. Column name for method 2 (y-axis).
--conf <value>	Optional. Confidence level for slope and intercept CIs (default: 0.95).

Example

```
jrc_corr_passing_bablok ~/data/method_comp.csv --xcol method_A --ycol method_B
```

What to look for in the output

The script tests two hypotheses. First, the proportionality test: the 95% CI for the slope should include 1 (no proportional difference between methods). Second, the systematic bias test: the 95% CI for the intercept should include 0 (no constant offset). If both CIs include their target value, the methods can be considered interchangeable over the tested range. The Cusum linearity test checks that the relationship is linear — if it rejects, the single-line model may be misleading. The PNG shows the data with the P-B line (solid blue), its 95% CI (dashed blue), and the line of identity y = x (dashed grey) for visual reference.

TIP: For a complete method comparison record in your DHF, run both jrc_corr_passing_bablok (regression + interchangeability tests) and jrc_bland_altman (agreement limits). These two analyses are complementary: Passing-Bablok shows the systematic relationship; Bland-Altman shows how large the individual differences are.

5. Preparing Your Data

All four scripts expect a CSV file with at least two numeric columns. Use --xcol and --ycol to specify the column names if they differ from the defaults.

Script	Required columns	Optional	Default --conf
jrc_corr_pearson	x, y (configurable)	—	0.95
jrc_corr_spearman	x, y (configurable)	—	n/a
jrc_corr_regression	x, y (configurable)	—	0.95
jrc_corr_passing_bablok	x, y (configurable)	—	0.95

Column names are case-sensitive. Rows with missing values in either column are automatically dropped before analysis. At least 3 complete paired observations are required; the script stops with an error if this minimum is not met. Extra columns in the CSV are ignored.

INFO: Your CSV does not need to use 'x' and 'y' as column headers. Any column name works. For example, if your file has columns 'method_A' and 'method_B', run:

```
jrc_corr_passing_bablok mydata.csv --xcol method_A --ycol method_B
```

6. Recommended Workflow

6.1 Exploring a new relationship

When you first want to understand the relationship between two continuous variables:

1. Run jrc_corr_pearson to get r, its CI, and a quick scatter plot. This confirms whether a linear association exists and how strong it is.
2. If the scatter plot looks non-linear, or if normality is in doubt, run jrc_corr_spearman as a check. If both r and ρ are similar in magnitude, the relationship is well-described as linear.
3. If you need a fitted line (calibration, predictive model), run jrc_corr_regression. Check the residual plot — if residuals are random, the linear model is valid.

6.2 Method comparison study

When qualifying a new measurement method against an existing one:

4. Run jrc_corr_passing_bablok. Check that the slope CI includes 1 and the intercept CI includes 0. If both pass, the methods are interchangeable.
5. Run jrc_bland_altman (community scripts) to quantify the limits of agreement. Confirm that the limits are clinically / technically acceptable for your application.
6. Document both PNG outputs and the numerical results in your Measurement System Qualification record in the Design History File.

WARNING: Do not use OLS regression (jrc_corr_regression) for method comparison if both instruments have measurement error. OLS assumes the x variable is measured without error, which biases the slope estimate toward zero. Use jrc_corr_passing_bablok instead, which makes no assumption about which method is the reference.

7. Glossary

Term	Definition
Pearson r	The product-moment correlation coefficient. Measures linear association between −1 (perfect negative) and +1 (perfect positive). Assumes normality.
Spearman ρ (rho)	The rank-based correlation coefficient. Measures monotonic association. Does not assume normality or linearity.
Fisher z	A transformation of Pearson r used to construct the confidence interval. The CI is computed in z-space then back-transformed to the r scale.
CI	Confidence interval. The range of plausible values for a parameter. A 95% CI means that if the study were repeated many times, 95% of such intervals would contain the true value.
R^2	Coefficient of determination. The proportion of variation in y explained by the linear model. $R^2 = 1$ is a perfect fit; $R^2 = 0$ means x explains nothing.
Residual	The difference between an observed y value and the value predicted by the regression line. Residuals should be randomly distributed around zero.
Passing-Bablok	A robust regression for method comparison that does not assume a gold-standard reference. Uses the median of all pairwise slopes as the slope estimate.
Cusum test	Cumulative sum of residual signs, used to test whether the Passing-Bablok line is a valid linear summary of the data. A significant result suggests the relationship is non-linear.
Line of identity	The line y = x, shown as a dashed grey reference in the Passing-Bablok plot. If the P-B line coincides with the identity line, the methods are perfectly interchangeable.
OLS	Ordinary Least Squares. The standard regression method that minimises the sum of squared vertical residuals. Assumes x is

	measured without error.
p-value	The probability of observing a result as extreme as this one if there were truly no association. $p < 0.05$ is the conventional significance threshold.
DHF	Design History File. The regulated record of all design and development activities for a medical device (21 CFR 820.30(j)).